CAPITAL ONE: MIGRATION TO AWS

IT457 – Cloud Computing Case Study | Week 1 Summary
Group 41

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1. Case Study Overview

Capital One, a US financial services company, adopted a cloud-first strategy to modernize its IT infrastructure and accelerate digital banking innovation. In 2020, it completed its exit from eight on-premises data centers to Amazon Web Services (AWS). The transformation involved infrastructure migration as well as application modernization and changes to software development and operations. [1]

2. Problem and Motivation

Traditional data centers required physical infrastructure management and resource provisioning. Capital One sought greater operational flexibility, faster development, improved resilience, and the ability for technology teams to focus more on customer-facing services. [1, 3]

3. Architecture and Migration Approach

The case study identifies AWS services such as Amazon EC2 (compute), Amazon S3 (object storage), and Amazon RDS (managed relational databases). In a documented customer-facing application migration, Capital One used Amazon Route 53 for traffic management, Auto Scaling Groups for capacity management, and blue/green deployment practices. Traffic was introduced gradually and the deployment was later extended to a second AWS Region. This example should not be assumed to represent every Capital One workload. [1, 2]

4. Reported Outcomes

- Exited eight on-premises data centers.
- Average development-environment build time fell from approximately three months to minutes.
- Disaster-recovery time in tests decreased by 70%.
- Critical incident resolution time and transaction errors each decreased by 50%.

5. Cost and Business Impact

Cloud adoption enabled more flexible provisioning and reduced reliance on physical data-center infrastructure. However, the total financial cost of the migration and overall net savings were not established in the sources reviewed. A quantitative cost comparison requires additional verified financial data.

6. Key Lessons

The case illustrates that cloud migration is more than moving servers: application redesign, phased traffic migration, automation, resilience planning, and cloud engineering practices are important. The proposed practical exercise is to move a Dockerized application from a local environment to a cloud VM, refactor one component to a managed service, and document effort, performance, and estimated cost.

Sources

[1] AWS, *Capital One case study*: https://aws.amazon.com/solutions/case-studies/capital-one-all-in-on-aws/

[2] Capital One Tech, *Moving a Customer-Facing App to AWS*: https://www.capitalone.com/tech/cloud/moving-customer-facing-apps-to-aws/

[3] Amazon, *Capital One Selects AWS as Its Predominant Cloud Infrastructure Provider*: https://press.aboutamazon.com/2016/11/capital-one-selects-aws-as-its-predominant-cloud-infrastructure-provider

Note: Operational metrics are reported in AWS's published case study and are not presented as independently audited results. Architecture details from [2] describe one application migration, not necessarily every Capital One workload.